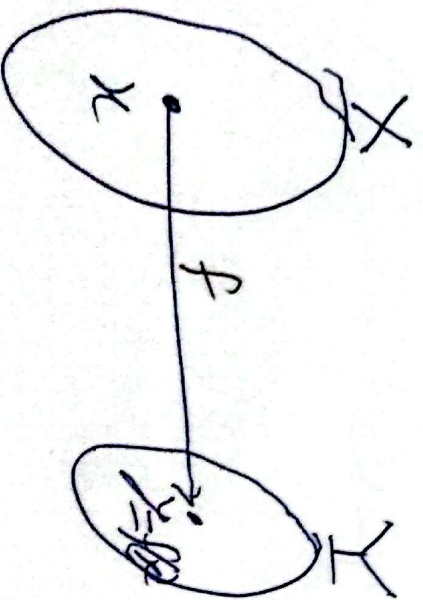


Function: - If X and Y are two non-empty sets and f is such a rule that gives a unique value $y \in Y$ for $x \in X$ then f is called a function ~~from~~ from the set X to the set Y . It is denoted by $f: X \rightarrow Y$.



Domain: - If $y = f(x)$ be a function then the set which contain the values of x , is called domain.

Range: - If $y = f(x)$ be a function then the set which contain the values of y for every values of x , is called range or codomain.

One-one function: - If $f(a_1) = f(a_2)$

$\Rightarrow a_1 = a_2$ then $f: A \rightarrow B$ is

called one-one function, where $a_1, a_2 \in A$.

onto function: -

DaykoF

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Inverse function: - ~~If $f: A \rightarrow B$~~
If $f: A \rightarrow B$ is a one-one
and onto function then f is
an inverse function.

Bounded function: - If the functions
 $f(x)$ is defined and values of it ~~exist~~
exist in the interval (a, b) and if
two real numbers m, M exist
such that $m \leq f(x) \leq M, \forall x \in (a, b)$
then $f(x)$ is called bounded function.

Decreasing function: - If $f(x)$ increases for the increasing values of x in any interval then the function $f(x)$ is called increasing function i.e. $x_1 > x_2 \Rightarrow f(x_1) > f(x_2)$.
e.g. $f(x) = x^3$

Decreasing function: - If $f(x)$ decreases for the increasing values of x in any interval then $f(x)$ is called decreasing function i.e. $x_1 > x_2 \Rightarrow f(x_1) < f(x_2)$.
e.g. $f(x) = \frac{1}{x^2}$

Even function: - A function $f(x)$ is said to be even if $f(-x) = f(x)$.

e.g. $f(x) = \cos x$

Odd function: - A function $f(x)$ is said to be odd if $f(-x) = -f(x)$.

e.g. $f(x) = \sin x$

Periodic function: - A ~~the~~ function $f(x)$ is called periodic function in its domain D if a positive number T exist in such a way that $f(x+T) = f(x)$, $\forall x \in D$.

Here T is called period.